Report- SE-DC Motor Control

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***Abstract*—The correct control of the SE DC Motor is crucially important in ensuring a smooth behaviour of the vehicle. This study sought to design a control system for a tramway such that it would provide a smooth ride for the passengers of the tram in cases of different speed and slope condition, specifically not focusing on the mechanical part involved in the analysis of the system, but focusing on the electrical part of the motor, including the limitation related to the operating regions of the component.**

For the detailed graphs,
please refer to the end of the document

I. Introduction

A Separately excited DC (SE DC) motor is used to move an ATM tramway vehicle "Carelli 1928". We list the provided characteristics of the system, using the symbol nomenclature to be followed in the present report:

Line voltage (nominal voltage)	$V_n = 600V$
Efficiency	$\eta = 0.9$
Armature circuit time constant	$\tau_a = 10ms$
Excitation circuit rated voltage	$V_e = 120V$
Excitation circuit rated current	$I_e = 1A$
Excitation circuit time constant	$\tau_e = 1s$

TABLE I: SE DC Motor Parameters

Additional information: The tramway should be able to accelerate from 0 to 60 km/h in 25 s. The tramway mass is 10000 kg and you should consider 200 people (with a standard weight of 80 kg) as the tramway trainload, since the analysis of stability in the worst case scenario would provide a robust control system even at lower masses. The friction force is proportional to the speed and at rated speed ($v_n = 60\ km/h$ or $\omega_n = 314\ rad/s$) is $\frac{1}{3}$ of traction force.

Required Task: Find the design parameters of the motor according to the data. Then, implement a control system to cover a 10 km track considering the Table II characteristics. The slope s is defined as $s\% = 100\ tan(\gamma)$, where γ is the angle of the slope with respect to the horizontal ground reference.

II. Modelling of the physical system

We show in Fig. 1 the control scheme for a SE DC motor discussed in class. Each of the subsystems (highlighted with different colors) will be discussed in the following subsections of the report.

Position range (km)	Slope (%)	Speed (km/h)
0 to 1	0	35
1 to 3	0	60
3 to 4	+5	60
4 to 6	0	75
6 to 8	0	60
8 to 9	−5	60
9 to 10	0	35

TABLE II: Kinematic characteristics of the trajectory to be followed by the tramway.

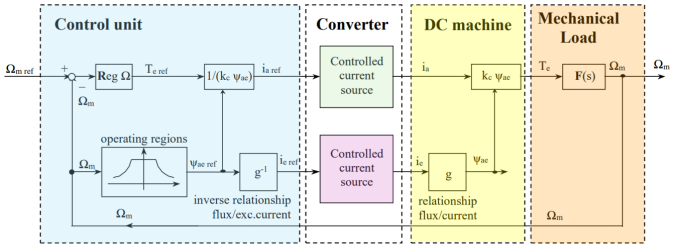


Fig. 1: Speed control scheme of a separately excited DC machine

A. Electrical modelling - Separately Excited DC Motor

We present on Fig. 2 the simulation diagram of a Permanent-Magnet DC (PM DC) motor. Considering that the voltage reference v^* is within the capabilities of the power supply, we will get that the actual voltage v applied to the plant is $v = v^*$. Furthermore, notice that the input given to the transfer function $G_i(s) = \frac{1}{R+sL}$ is $v - e$, where $e = K\omega$ is the back electromotive force (back-EMF). In addition, observe that the output of the transfer function $G_\omega(s) = \frac{1}{\beta+sJ}$ is ω, which is a signal used to compute e. Therefore, those two transfer functions are indeed coupled by the term $e = K\omega$.In order to simplify the control design and disentangle the dynamics of the two transfer functions mentioned before, we add a feed-forward term $\hat{e} = K\hat{\omega}$ to the voltage reference v^*, in which hat symbol indicates that we are either estimating or sensing those variables. Now, if we compute the input of $G_i(s)$, we obtain $v + \hat{e} - e$. Provided that we can execute a good back-EMF estimation ($\hat{e} = e$) and considering that the voltage supply works properly ($\hat{v} = v$), we obtain a signal $v + \hat{e} - e = \hat{v}$ that does not depend on either e nor ω and allows us to decouple the dynamics of the transfer functions $G_i(s)$ and $G_\omega(s)$. To convert this PM DC dynamical model to the corresponding one for a SE DC motor, we need to do the following changes: $R \to R_a$, $L \to La$, and $K \to K\ I_e$. We can then apply the same concept of using the back-EMF compensation for decoupling the speed control and the (armature) current control.

B. Mechanical Modeling - The Tramway Model

The analysis of the mechanical components is not reported here due to a lack of interest in this components with respect to the electrical modeling, which is the actual focus of the whole experiment.

C. Model Representation - SimuLink Block Scheme

For the Simulink Block Scheme please refer to the last page of the document.

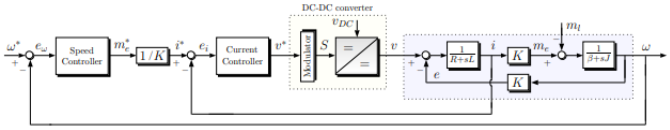


Fig. 2: Simulation diagram of a Permanent-Magnet DC motor

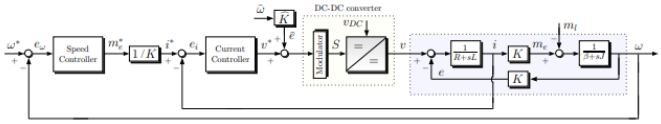


Fig. 3: Simulation diagram of a Permanent-Magnet DC motor, with back-EMF compensation

D. Parameter Identification

1) From the problem definition (see Section I): Total mass of the vehicle loaded with all the 200 passengers:

$$M = 10 * 103 + 200 * 80 = 26000\ Kg \tag{1}$$

2) Given $v_n(in\ \frac{km}{h})$ and t_{req}, from the problem definition (see Section I): Average vehicle acceleration:

$$a_{avg} = \frac{\Delta v}{\Delta t} = \frac{v_n}{t_{req}\ *\ 3.6} =\ 0.667\ m/s \tag{2}$$

3) Given M and a_{avg}, from Eqs. (1) and (2): Traction force:

$$F_t = M\ a_{avg} = 17.333\ kN \tag{3}$$

4) Given F_t and v_n, from Eq. (3) and the problem definition (see Section I): Traction power of the vehicle:

$$P_t = F_t\ v_n = 288.889\ kW \tag{4}$$

5) Given Pt, from Eq. (4) and the problem definition (see Section I): Friction power for the vehicle:

$$P_f = \frac{P_t}{3} = 96.296\ kW \tag{5}$$

6) Given Pt and Pf , from Eqs. (4) and (5): Nominal mechanical power:

$$P_{N,m} = P_t + P_f = 385.185\ kW \tag{6}$$

7) From the definition of efficiency in a SE DC motor, and considering the approximation $V_n I_n \gg V_e I_e$:

$$\eta = \frac{P_{N,m}}{V_n\ I_n + V_e\ I_e} \Rightarrow \eta \approx \frac{P_n}{V_n\ I_n} \tag{7}$$

8) Given P_n and η, from Eqs. (6) and (7) and the problem definition (see Section I): Nominal current:

$$I_n \approx \frac{P_n}{V_n\ \eta} \Rightarrow I_n \approx 713.306\ A \tag{8}$$

9) Given P_n and ω_n, from Eq. (6) and the problem definition (see Section I): Nominal torque:

$$T_n = \frac{P_n}{\omega_n} = 1226.7\ Nm \tag{9}$$

10) Given I_n and T_n from Eqs. (8) and (9), and assuming $K(Ie) = K\ I_e$:

$$K = \frac{T_n}{I_n\ I_e} = 1.7197\ \frac{Nm}{A^2} \tag{10}$$

11) Power dissipation at the armature circuit:

$$Pa = R_a\ {I_n}^2 \tag{11}$$

12) Given P_n, P_a, V_n, and I_n, from Eqs. (6), (8) and (11) and the prob. def. (see Section I): Power conservation:

$$\begin{aligned} P_n + P_a = V_n\ I_n \Rightarrow \\ P_n + R_a\ {I_n}^2 = V_n\ I_n \Rightarrow \\ R_a = 0.0841\ \Omega \end{aligned} \tag{12}$$

13) Given Ra and τa, from Eq. (12) and the problem definition (see Section I): Armature circuit inductance:

$$L_a = R_a\ \tau_a = 841\ \mu H \tag{13}$$

14) Given V_e and I_e, from the problem definition (see Section I): Armature circuit voltage:

$$V_e = R_e\ I_e \Rightarrow R_e = 120\ \Omega \tag{14}$$

15) Given R_e and τ_e, from Eq. (14) and the problem definition (see Section I): Excitation circuit inductance:

$$L_e = R_e\ \tau_e = 120\ H \tag{15}$$

16) Given M, v_n, and ω_n, from Eq. (1) and the problem def. (see Section I): Balance of Kinetic Energies:

$$\begin{aligned} \frac{1}{2} M\ {v_n}^2 = \frac{1}{2} J_{eq}\ {\omega_n}^2 \Rightarrow \\ J_{eq} = 73.2507\ Kgm^2 \end{aligned} \tag{16}$$

17) Given P_f and ω_n, from Eq. (5) and the problem definition (see Section I): Torque loss from shaft friction:

$$T_f = \frac{P_f}{\omega_n} = 306.68\ Nm \tag{17}$$

18) Given T_f and ω_n, from Eq. (17) and the problem definition (see Section I): Friction coefficient β (assuming that the friction torque at the shaft is proportional to the angular speed):

$$\beta = \frac{T_f}{\omega_n} = 0.9767\ Nms \tag{18}$$

19) Given η and V_n from the problem definition (see Section I), and using Eq. (7): Rated Back-EMF:

$$E_n\, I_n = \eta\, V_n\, I_n \approx P_n \Rightarrow$$
$$E_n \approx \eta\, V_n \Rightarrow$$
$$E_n \approx 540\, V \quad (19)$$

20) Given v_n and ω_b from the problem definition (see Section I): Radius:

$$r = \frac{v_n}{\omega_b} = 0.0531\, m \quad (20)$$

III. Control design for the physical system

In this section we describe the control system designed and how we tuned the corresponding parameters.

A. Control description: PI controllers in cascade

As previously presented in Fig.1 and then detailed in the SimuLink scheme, we have a cascade control system with three controllers: the armature current controller (shown in green) that is nested within angular speed control loop (shown in light blue), and the excitation current controller (not in cascade, and shown in pink). For each one of them, we used a PI controller (Proportional and Integral) and in some cases we introduce a saturation. We didn't include a derivative action for two main reasons: one is related to the actual proprieties imposed by the derivative, the term improves the response overshoot and we will design the transient response to avoid it (by setting the robustness to the highest value supported, which means having a phase margin of $90°$), while the second reason is related to a realization problem of the derivative component. Also, by choosing PI instead of PID controllers we reduce the complexity of the design. The tuning parameters were configured according to Table III. Modifying the response time (τ_r) changes only the bandwidth ($f_{BW} \equiv f_{3dB}$), but not the phase margin. In essence, with a lower time response, we obtain a higher bandwidth. Recalling that $\tau_r \approx \frac{0.35}{f_{3dB}^{(inHz)}}$ (with f_3dB given in Hz) and $f^{(inrad/s)} = 2\pi f^{(inHz)}$, we can obtain the following relation between the response time and the bandwidth given by the PID Tuner: $\tau_r \approx \frac{0.7\pi}{f_{3dB}^{(inrad/s)}}$.

Controller	Response Time (s)	Phase Margin ($°$)
Excitation current	$t_e = 0.075$	90
Armature current	$t_a = 0.1$	90
Angular Speed	$t_\omega = 1$	90

TABLE III: Controller parameters given to the PID Tuner tool.

In the same way, the transient behaviour (TB) is more aggressive (lower coefficient) when we reduce the phase margin ϕ_m. For this pair of coefficients, the relation is: $TB = \frac{\phi_m}{10}$, with ϕ_m in degrees. Recall that we are in a nested control loop. There are cascade constraints that were considered, regarding the response times: the outer (angular speed) loop must be at least ten times slower than the inner (armature current) one. Regarding the phase margin, we chose a phase margin of $90°$, as discussed in the beginning of this subsection. These two design parameters (see Table III) were provided to the PID Tuner tool, as shown in Figure 4. Notice that $\tau_a = \frac{\tau_\omega}{10}$, and recall that the excitation current control loop is not nested with the other ones.

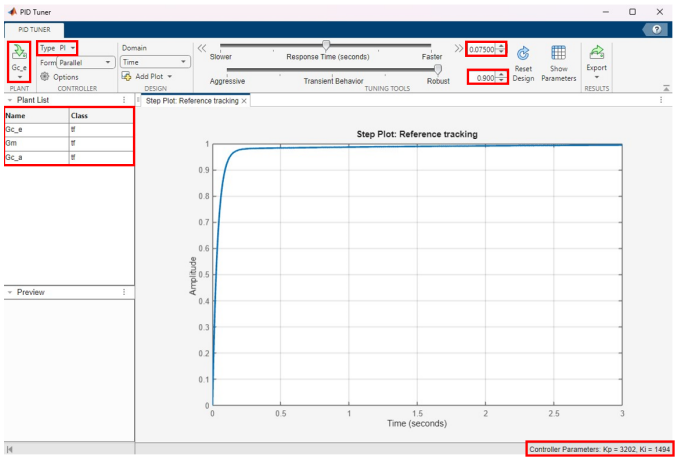


Fig. 4: PID Tuner visualization

With those parameters, the MATLAB PID Tuner calculates the bandwidth, transient behavior coefficients, and the PI controller gains (K_p and K_i) for the three controlled systems. The parameters obtained by the software are shown in Table IV.

Controller	Bandwidth ($\frac{rad}{s}$)	TB	K_p	K_i
Excitation current	26.9	0.9	3202	1494
Armature current	20	0.9	0.01682	1.682
Angular Speed	4	0.9	296	7.691

TABLE IV: Controller parameters obtained by the PID Tuner tool.

After that, we exported the controller variables to the MATLAB workspace. We then saved the controllers with the command: save('$PIDs_ready.mat$','C'_e','C'_a','C'_s'). Hence, we had the tuner controllers accessible from an external file.

B. Problem: Constraints of physical actuators

Even when modelling the system as if it was linear, there are some constraints that need to be considered to improve the performance and robustness of our control system. For instance, in the physical world the actuators have limited capabilities regarding the range of possible values that they can apply to the plant, and/or the speed in which they can change their action produced. This non-linear effect is called saturation. On the other hand, during the design we could intentionally impose saturation restrictions for the actuators, in order to protect the electronics or other pieces of hardware.

C. Dealing with plant constraints: Saturation and Windup

However, when the reference signal reaches its saturation state, the integral component of the control signal will continue to increase even if that extra accumulation doesn't affect the output of the plant. This undesired behaviour is known as *windup*.

D. Anti-Windup Strategy: Back-calculation

Literature wise, the are several strategies that can be used to address the windup problem: conditional integration, incremental algorithm, observer approach, gain scheduling and back-calculation. The latter method, whose block diagram is shown in Figure 5, was used in the project. To understand this procedure, observe that the anti-windup signal is negative when $v > u$, which allows to "discharge" the integral component of the original controller. Besides that, notice that the anti-windup signal is zero when there is no saturation ($v = u$), and the vanilla controller is unchanged. Also, notice that the back-calculation coefficient is $\frac{1}{T_t}$.

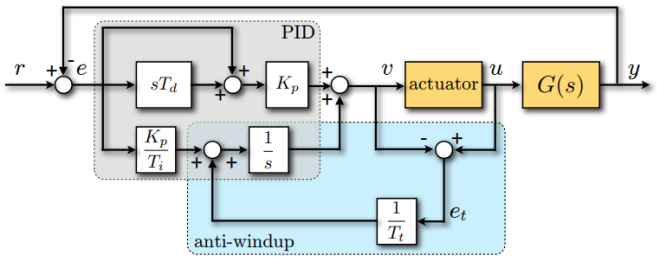


Fig. 5: Anti-Windup strategy based on back-calculation.

Implementing this scheme in the simulation is quite simple, given that the PI controller already contains this function. Indeed we configured the *Saturation* on the physical limits of the system (i.e. the limits of the operating region on the armature voltage) and we defined the saturation limits, the anti-windup strategy, and the back-calculation coefficient (see Figure 5). For the two current controllers, the saturation limit is $V_{line} = 600\,V$ because they generate voltage control signals that need to be taken from the available power supply (see Section I).

IV. Results and Discussion

In this section, we present the results of the simulations executed using MATLAB and Simulink. The numerical solver used is $ode23s$, with internal parameters left as default.

A. Verification of kinematic requirements

The tramway must cover a $10\ km$ track considering the Table II characteristics. The simulation time (on the horizontal axis) was $800\,s$ for the following two experiments. As we can see in figure 6, the requirements of Table II are well respected after the Transient period, and clearly at every change in value, there isn't an impulsive response, but a much more moderate behaviour. In the picture, the graphs are depicted as follows:

- linear speed reference (in red),
- simulated linear speed (in blue)
- road slope (in orange).

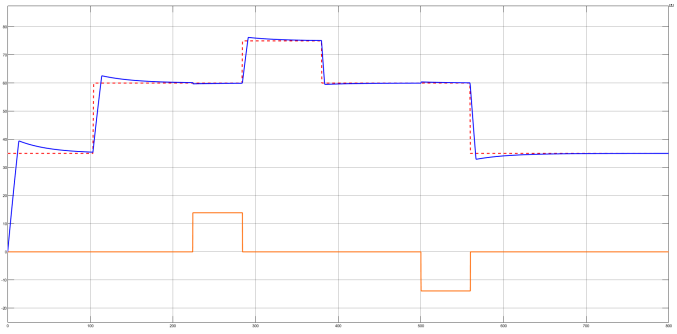


Fig. 6: Verification of the requested kinematic characteristic

B. Excitation current control

In Figure 7, we can observe the variation of the excitation current according to the operating regions. When the DC motor enters the constant power zone (field weakening region) by surpassing the base speed ($\omega_n = 314\ rad/s$), the excitation current (also called field current) becomes lower than its nominal value. The graphs are depicted as follows:

- excitation current reference (in blue),
- simulated excitation current (in red)

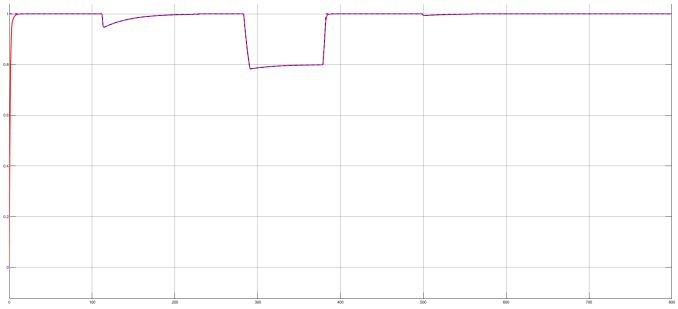


Fig. 7: Excitation current control

In addition we can also refer to the following graph to showcase the error on the excitation current according to the equation:

$$\epsilon_{i_e} = \frac{Ie - Ie_r}{Ie_r} \tag{21}$$

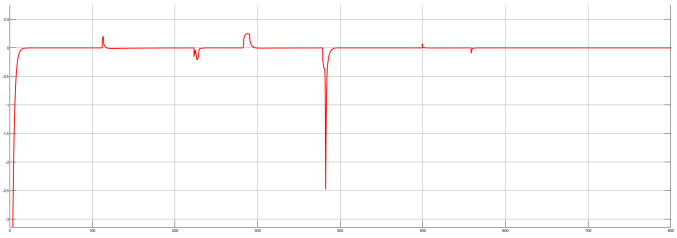


Fig. 8: Excitation Current Error

As we can see, the results are expressed in percentage points and, if we do not consider the initial transient up to about 20 secs, we have a very low error ranging between -2.5% and 0.5%

C. Armature current control

In Figure 9, we started the reference signal of the armature current $10\ s$ after initiated the simulation (as shown in the blue region within of the diagram of Figure **??** in Appendix VII). This delay is implemented to allow the excitation current circuit to reach its steady-state. Also, we can see the presence of pulses on the armature current reference to push the angular speed up when there's a change in the required speed of the tramway. On the periods when the required angular speed is constant, and there's no external load perturbation, the armature current converges to an stable value. Finally, we can see the effect of the anti-windup strategy implemented in the controller (for example, on the time interval $t \in [200\,s; 500\,s]$. The graphs are depicted as follows:

- armature current reference (in blue),
- simulated armature current (in red)

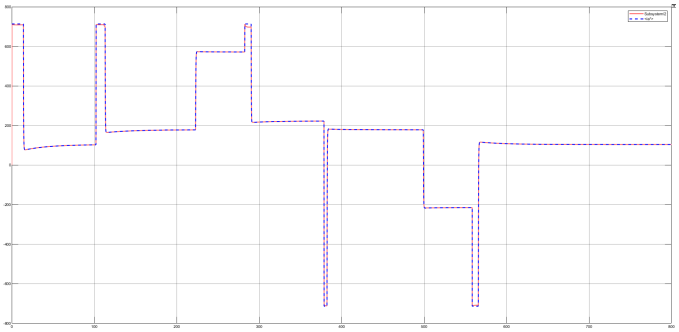


Fig. 9: Armature current control

D. Angular speed control

As depicted in Figure 10, the settling time obtained is a bit higher compared with the other control loops (which means a much slower response), but it allows us to not have high overshoots on the transient state of the angular speed. Also, we can see that the tramway follows the desired reference, but it presents some overshoots and undershoots on the time instants where there are changes on the reference speed or the external load m_l.

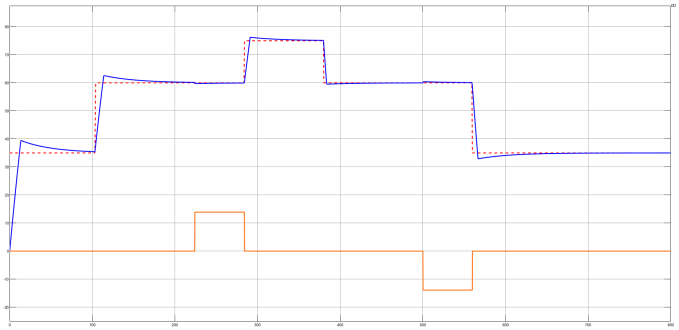


Fig. 10: Verification of the requested kinematic characteristic

E. Torque at the motor and external torque load

In Figure 11 we can see the torque reference signal m_e (in red), and external torque m_l (in blue). When the reference is saturated, we can see the effect of the anti-windup scheme. Also, we can see the changes in the external load m_l, which correspond to the track positions where there are changes in the slope (as requested in Section I). Besides that, we can see negative reference torque when we want to reduce the speed of the tramway (see timestamps $400\,s$ and $600\,s$). Also, in zones where the angular speed is constant (see near timestamp $500s$), the torque me decreases when the external load m_l decreases. To understand that, notice that $m_e - m_l = \beta\omega$: constant (recall $m_e - m_l - \beta\omega = \tilde{J}\,\dot{\omega}$ and $\dot{\omega} = 0$).

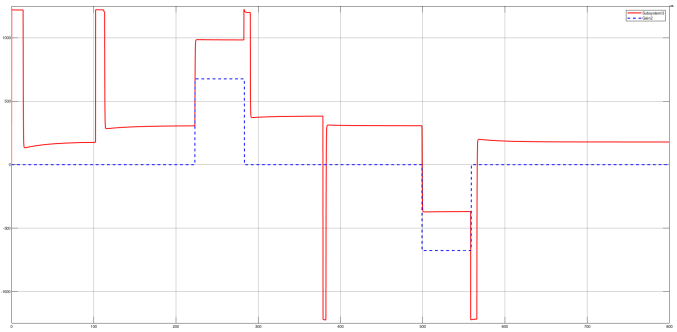


Fig. 11: Torque at the motor and External Torque Load

V. Conclusion

In this report, we managed to model a SE DC machine using MATLAB and Simulink, and then analyzed the obtained results verifying the theoretical relations discussed in the classes. Also, we used the flux weakening strategy for the angular speed control of the motor, and checked the expected behavior of the resulting operating regions.

VI. Appendix A
MATLAB script

```
M = 10*10^3+80*200;
g = 9.81;
Va_n = 600;
omega_b = 314;
eta = 0.9;
tau_a = 10^-2;
tau_e = 1;
Ve_n = 120;
Ie_n = 1;

Re = 120;
Le = 120;
dV = 60/3.6;
dT = 25;
r = dV/omega_b;

a_avg = dV/dT;
P_mecc = 4/3*M* a_avg*dV;
T_n = P_mecc / omega_b;
Ia_n = P_mecc/(eta*Va_n);
Ra = (Va_n * Ia_n-P_mecc)/Ia_n^2;
La = tau_a * Ra;
K = P_mecc/(omega_b*Ia_n);
E_n = eta*Va_n;
J = M *(dV/omega_b)^2;
beta = P_mecc/(4*omega_b^2);

vec_track = [0 2.999 3 3.999 4 7.999 8 8.999 9 10];
vec_slope = [0 0 0.05 0.05 0 0 -0.05 -0.05 0 0];
vec_track2 = [0 0.99 1 3.99 4 5.99 6 8.99 9 10];
vec_speed = [35 35 60 60 75 75 60 60 35 35];

kpi = 0.01682;
kii = 1.682;
kps = 148*2;
kis = 5.1274*1.5;
%kps = 1.739;
%kis = 0.02319;
kpe = 3202;
kie = 1494;

k_b_e = tau_e^-1;
k_b_a = tau_a^-1;
open("C:\Users\andre\Desktop\DEMD\PMDC\Simulink_DC_Motor_NOT_OPTIMIZED.slx");

plotoptions = bodeoptions;
plotoptions.Grid = 'on';
Sc_a = tf (1 , [La Ra]) * tf([kpi kii],[1 0]);%Gc_a*Rc_a

Sc_ = tf (1 , [J beta]) * tf([kps kis],[1 0]); %Gm*Rc_m


Sc_e = tf (1 , [Le Re])* tf([kpe kie],[1 0]);%Gc_e*Rc_e;
```

VII. Appendix B

Graphs

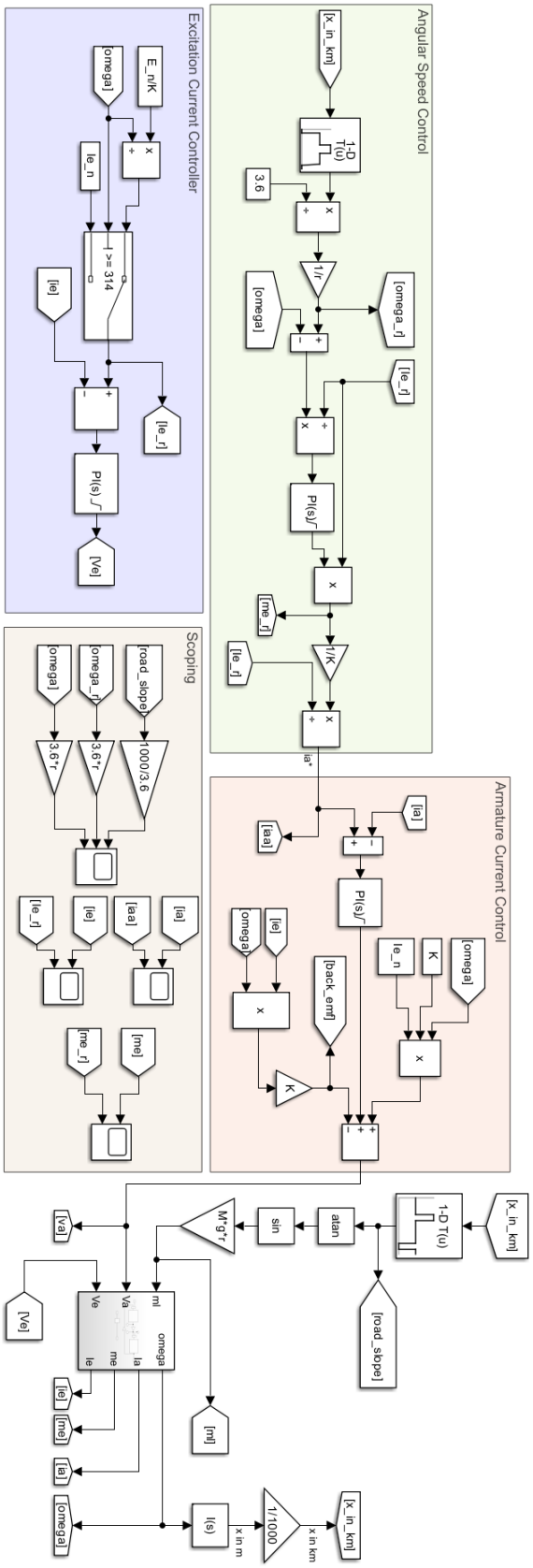
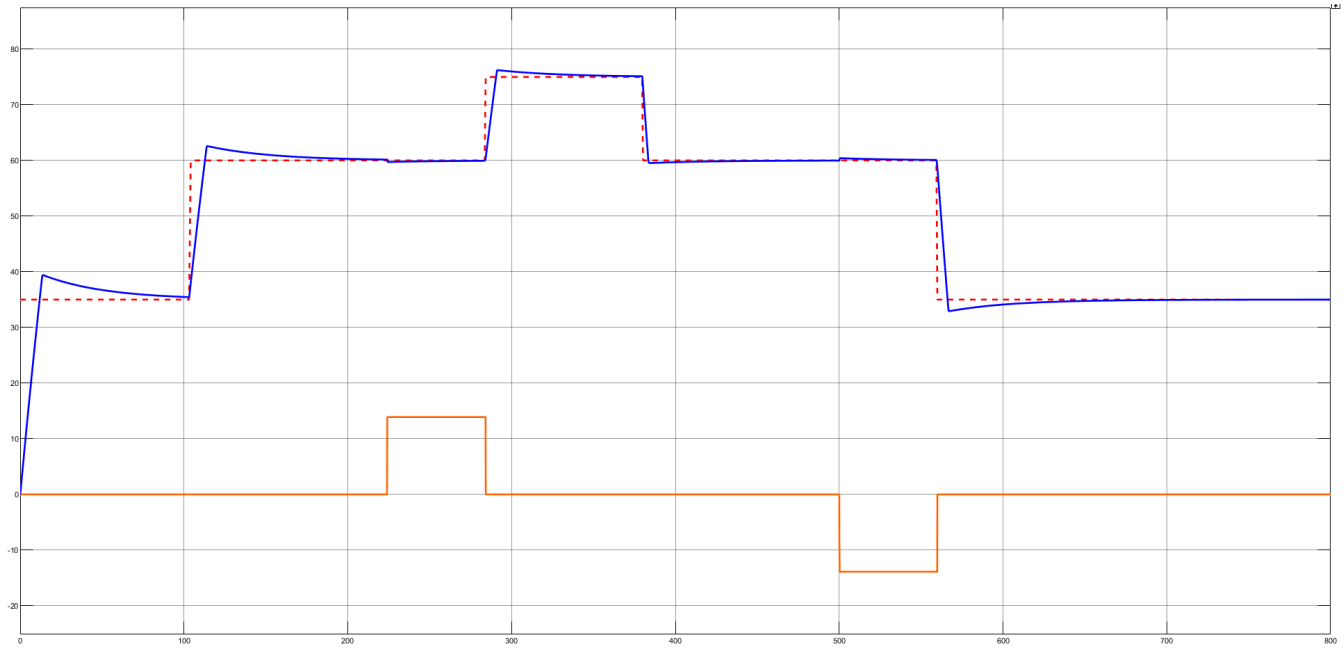
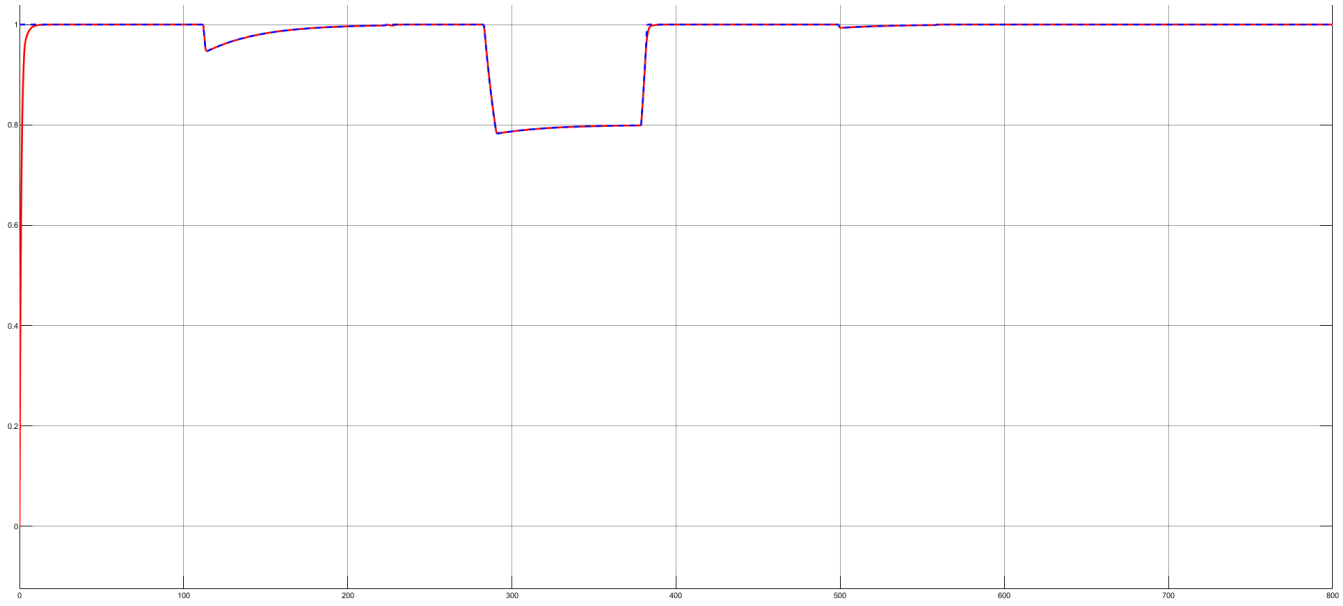


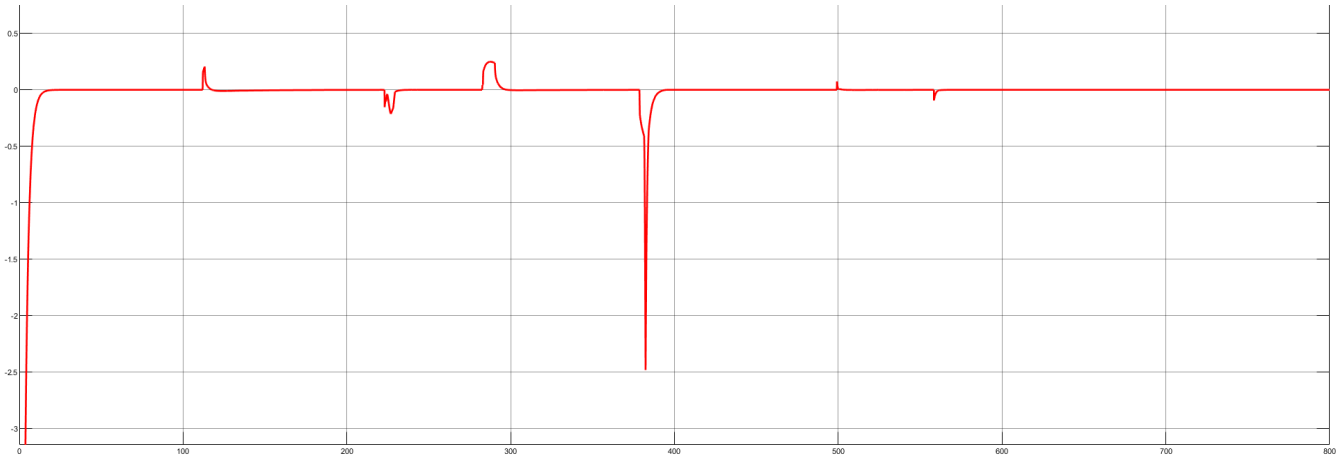
Fig. 12: Complete SimuLink Scheme



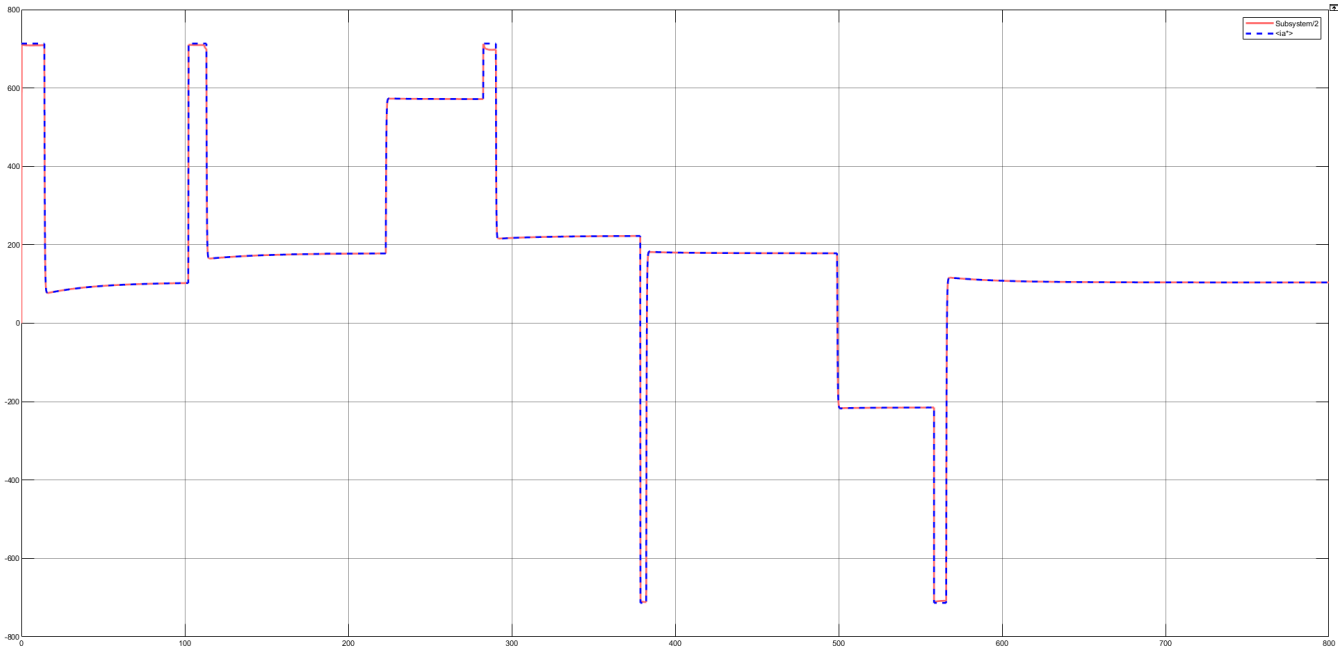
Verification of the requested kinematic characteristic



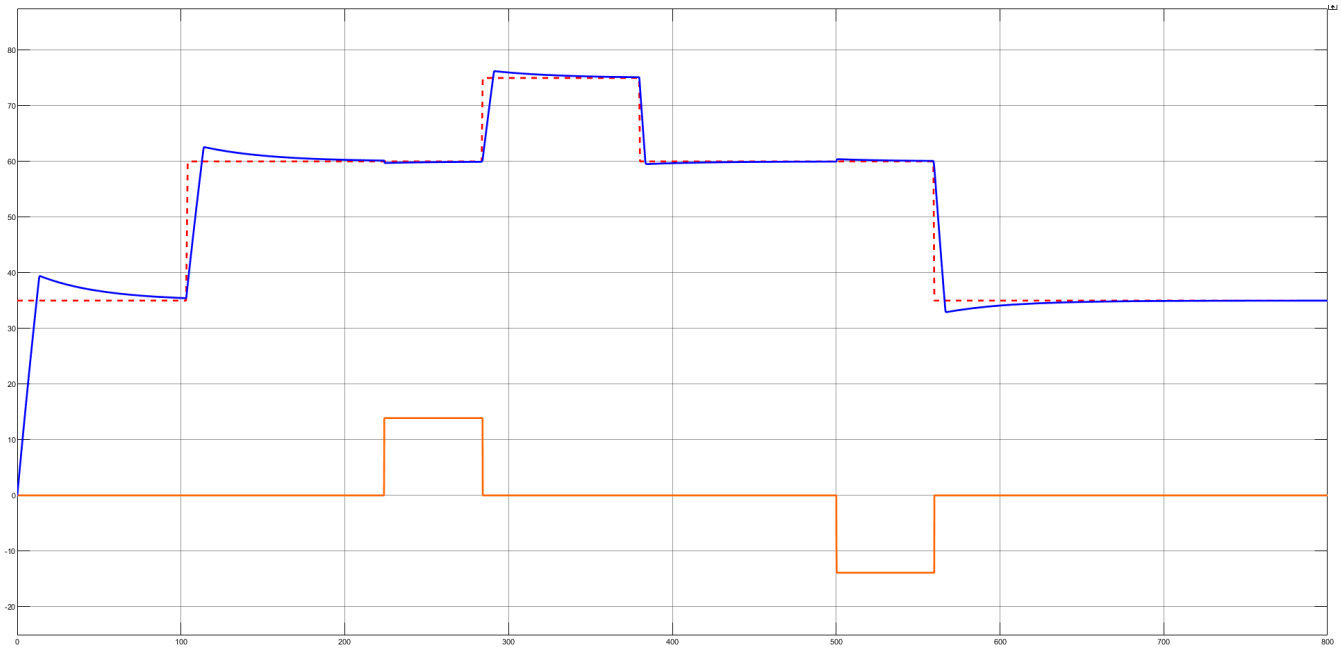
Excitation current control



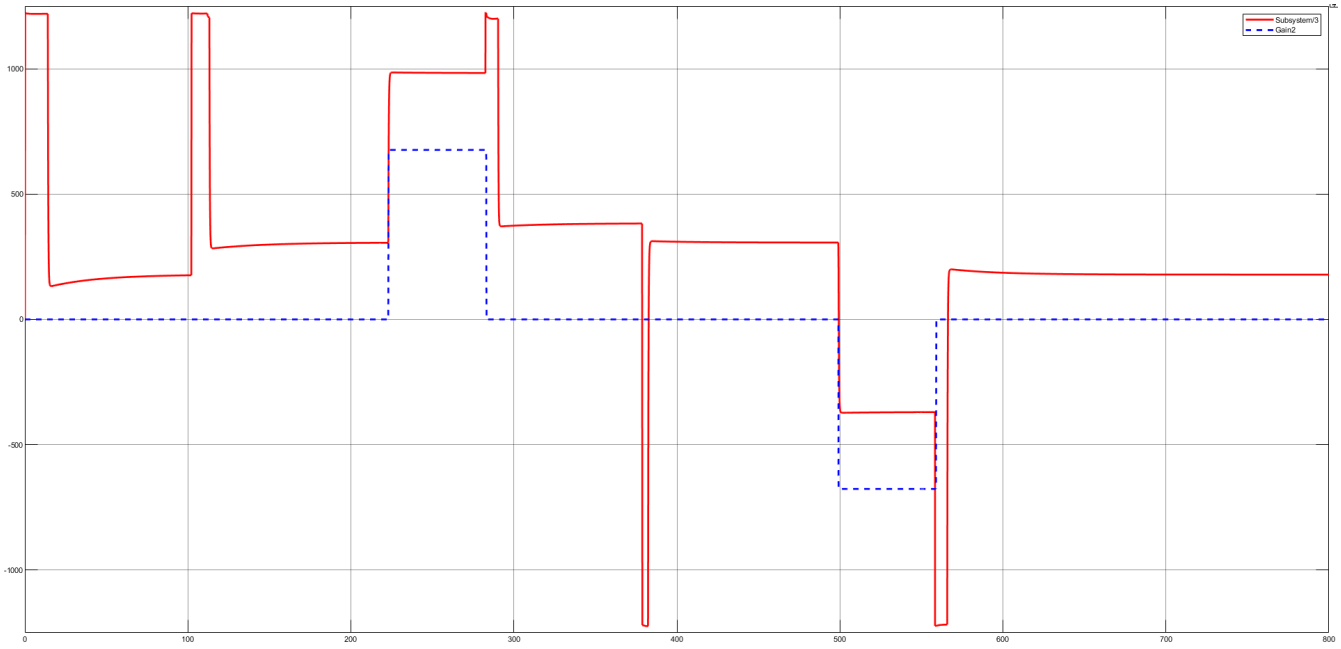
Excitation Current Error



Armature current control



Verification of the requested kinematic characteristic



Torque at the motor and External Torque Load